Possible Problems for Quiz 10 to be taken 4/4/19. Of these problems I will choose 3 to be on the quiz.

Problem 1

Let $\lambda_1 = 2, \vec{u}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\lambda_2 = -3, \vec{u} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$ be the eigen values and eigen vectors of a matrix A. Find the matrices D, P in the orghogonal diagonalization of A.

Problem 2

Let $A = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ calculate the condition number of A

Problem 3

Find the SVD of $A = \begin{bmatrix} 1 & 2 \\ 2 & 0 \\ 0 & 2 \end{bmatrix}$

Problem 4

Prove that for a real matrix A, AA^T has positive eigen values.

Problem 5

Let A, B, C be orthogonal matrices. What is $(ABC)(C^T B^T A^T)$ equal to? Justify your answer.

Problem 6

Find an orthonormal basis for $S = span\left\{\begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}\right\}$